

MAJOR UPDATE

We can now get the data in HTML format:

http://fts.unocha.org/pageloader.aspx?page=searchProject-reporting_display&CQ=cq240810065122BuluUkj270

Google Spreadsheet does a quick import:

<https://spreadsheets.google.com/ccc?key=0Asb9jTnW9nPsdE01c3BNR1M5MGFDX1hZRmxhbWdjcmc&hl=en&authkey=CMe11IoDff>

NEED: Professional level charting from the data. (see example below in document

Inserted comments/questions have moved to the "What has happened so far section" of Part 1.

Translating FTS data into dynamically created charts (and beyond)

FTS = [Financial Tracking System](#) (a global, real-time database which records all reported international humanitarian aid - [read more](#))

My initial thoughts are that this "project" would be done in two phases. The first would be to find a way to read FTS data into something like Google Docs dynamically with that data feeding into edynamically-created [charts for a specific emergency](#) which could be embedded onto any [website](#). The second phase would be "take a step higher" and produce a map of all FTS Flash/Consolidated Appeals and display it on a Google-like map where people can click on the icon and see the respective details/charts/etc.

For the Pakistan emergency, I am mostly interested in Part One. However, if anyone is keen to help find a solution, I want to make sure that whatever is designed in Part One can be easily leveraged to help in the Part Two.

Part 1 - Dynamic charting

Overall desire: dynamically pull Table D [[PDF](#) | [XLS](#)] from the [FTS data for the Pakistan Floods](#) and create [specific charts](#) (ideally within Google XLS) which can be embedded onto a website (like the [Finance page](#) on PakResponse).

Final product: some form of embeddable gadget that contains a [variety of charts](#) which show the funding level in the Pakistan Floods emergency. As you will see in the [existing image](#) that is displayed on the [financial page](#), the gadget would need to show several things:

1. Horizontal bar chart that shows the amount of funding required and the amount committed for each cluster
2. Small pie charts that show the percentage of funding for each cluster
3. A larger pie chart that shows overall funding requirements, funding committed, and corresponding numbers (\$ value of each, % of each, and \$ value of overall appeal)
4. Clear title, labels and legends.

Current data:

Until now, you might be thinking that this concept seems easy. Unfortunately, FTS does not provide their data in a nice, readable format

- [Pakistan Floods FTS page](#) (emergID=15913)
- Pakistan Floods - Table D: [XLS](#) | [PDF](#) (these are the only summarized form of the data)
 - Is there are a more raw form of these data? Or are you happy with the aggregate information - Tim McNamara 22/08/2010 10:00
 - Not that we can get at. I will begin negotiating with HQ (i.e. FTS), but not sure when/if that will work out.
 - However, for what we want to display, the aggregate information is completely fine.
- XML URL for the contributions to Pakistan:
<http://fts.unocha.org/ftswebservice/ftsfeed.aspx?datatype=emergency&id=15913>
 - Data is listed at the project level - if it can be summarized to find all the correct information/data, then it would have to be grouped on "sector".
 - To calculate the \$ figures, the value in the "usd-equivalent" attribute would need to be used (not the original currency amount)
 - FYI: I am not 100% sure if all the necessary data is in there to make it work....but if so, it might be easier than trying to figure out how to read the XLS and/or PDF docs....but perhaps slower if it has to read through all of the projects.

A couple specific Functional Requirements:

- The colors should be easily update-able - that is, I can change a color value in one XLS cell and the chart would be updated with the new color.
- References to the appeal, the URLs to data, the emergency/appeal ID, etc should not be 'hard coded' into the product --- again, they should reside in separate XLS cells which are then referenced
 - Hopefully, developing in this approach now will make using this in the next emergency significantly easier & simpler to roll into Part 2.

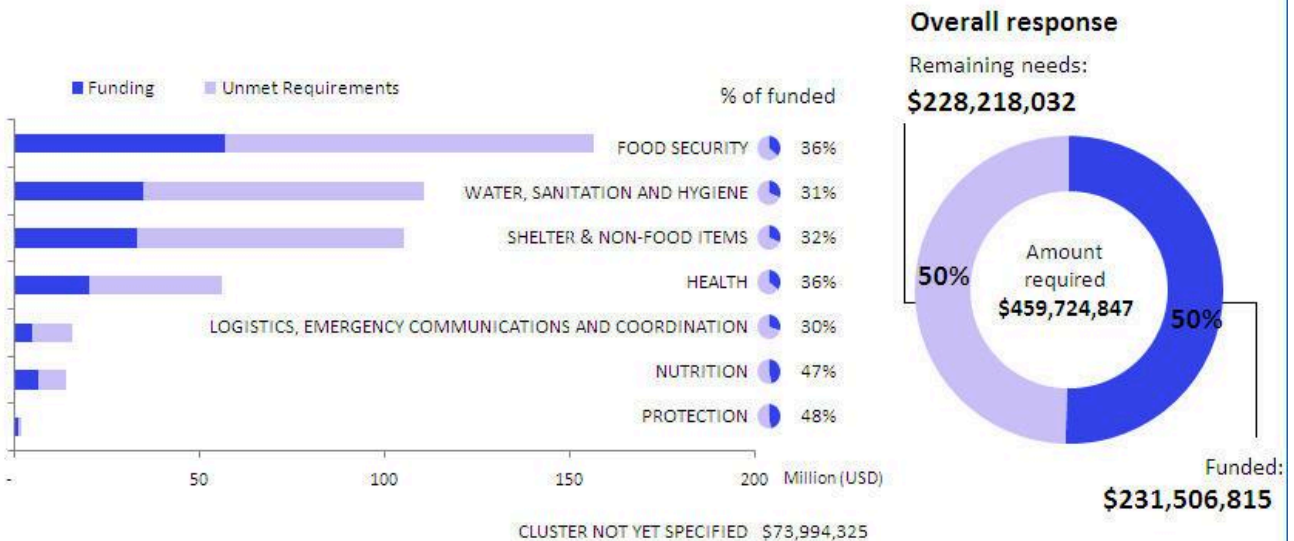
Basic Info from Google Spreadsheets/Charts:

- Google Chart Tools: <http://code.google.com/apis/charttools/index.html>
- Google Spreadsheet import functions:
 - <http://docs.google.com/support/bin/answer.py?answer=82712&query=Import>
 - as you can see, spreadsheets can pull in CSV, TSV, XML, HTML, and other feeds
 - More info: <http://docs.google.com/support/bin/answer.py?answer=75507>

Request:

- Initial Comment: Although I am most comfortable with Google Spreadsheets and the charting abilities, I am happy to consider any solution that is 'open', flexible, and can be replicated for future emergencies (without me having to ask someone each time). Whatever is developed must be extremely easy - so easy that a 'non-techie' can make it work.
- Develop a Google Spreadsheet that "automatically" imports FTS data for the Pakistan Floods Appeal and feeds this data into charts which can be embedded onto a website.
- Hopefully, my previous descriptions are enough to guide every/anyone.

HUMANITARIAN FUNDING AS AT 19 AUGUST 2010



What has happened so far:

- csadams@google.com played with Google Spreadsheet import
 - Not a straight forward import into Google Spreadsheet.
 - Looking into XPath Query
- @patrickmeier forwarded the concept to Eric Gundersen & Robert Soden of [Development Seed](#)
 - No response yet.
- @timClicks notes that [he is doing some work](#): <http://github.com/timClicks/reliefweb-fts>
- @google disaster and chart/visualization team aware of need
 - Perhaps put this into Trix first (there are programmatic ways to do this via [Andrej Verity](#))
 - You can upload a new version to the spreadsheet you created using the API and from there you can have the motion chart feeding off of the Trix sheet.
- guardian.co.uk seem to have FTS data in a Google Spreadsheet. Appears to be static, but worth looking at. Spreadsheet is then used to populate a java-based bubble chart.
 - [Guardian page](#) with chart and link to data
 - [Google Spreadsheet with data](#)
- @hillel - you can import a whole XLS file into google spreadsheet programatically using google data api
 - http://code.google.com/apis/documents/docs/3.0/developers_guide_java.html#UploadingDocContents

I tried to see what [google apps scripts](#) can offer but didn't see any code that can convert

an XLS file in one go ...

- @hillel - the current XLS format is full with empty rows/cols that makes the import result very bad. If the table structure of the XLS is kept the resulting google spreadsheet would be easier to use.
 - **Andrej:** yes - but I am not worried about the actual 'format' of the imported

XLS - we would just be using it as a reference sheet for data to populate the chart(s).

- @hillel - the set of charts available in any google spreadsheet consist of piechart, barchart etc. although not exactly of the type depicted above.
 - **Andrej**: understood. Also looking at Google Visualization tools & Google Chart (API).
- Chris Nicholas:
 - OpenOffice can be started i server mode and fully controlled via Java; one can extract the text via:
<http://pdfbox.apache.org/commandlineutilities/ExtractText.html>
 - OpenOffice server
install: <http://doc.nuxeo.org/5.3/books/nuxeo-book/html/admin-openoffice.html>
- Max Richman - It may be worth seeing what you can do clean up the data in Excel with macros, play with Microsoft's built-in pivot tables/charts, and then potentially explore how that can link to Microsoft Office 2010 online. (sorry Google peeps, just brainstorming out loud) :]
 - In Excel, you can probably clear out many of the unnecessary rows, columns, and merged cells:
 - Macro for removing empty columns:

```
Sub DeleteEmptyColumns()  
first = Selection.Column  
last = Selection.Columns(Selection.Columns.Count).Column  
For i = last To first Step -1  
If WorksheetFunction.CountBlank(ActiveSheet.Columns(i)) = 65536 Then  
Columns(i).Delete  
End If  
Next i  
End Sub
```
 - The other major problem is merged cells. Here are two ways to deal with this:
 - 1) One way to remove all merged cells in a sheet at once:
Select all cells in the sheet. Choose Format/Cells/Alignment. If there are merged cells, the Merge checkbox will have a horizontal line through it. Click it once to check it, then once more to clear it, then click OK.
(Source:
<http://forum.soft32.com/mac/Problem-merged-cells-ftopict83027.html>)
 - 2) A macro as follows:

```
Public Sub RemoveMergedCells()  
ActiveSheet.UsedRange.MergeCells = False  
End Sub
```
 - Running both of these could help clean up the excel tables and allow you to build a pivot table and perhaps more easily upload the table into a web-based table.
- Unfortunately even after attempting this type of clean-up in excel, the data do not align automatically and still require a few manual corrections.
- Found an [HTML link](#) via an OCHA colleague
- csadams@google.com setup the dynamic import into a [Google Spreadsheet](#)
- Patrick Tague (tague@cmu.edu) - As an alternate to using Google Spreadsheets, it would be pretty easy to create a web form that would ask the user to enter a web address (similar to those entered in the Google Spreadsheet) and a little info about the chart they want (which columns, what type of chart, etc.), use PHP to grab the data directly from the website, parse using PHP, and create charts using the Google Charts API. An example (with the web address hard-coded)that I threw together

during the CrisisCampSiliconValley event on 8/27/10 is found at <http://wnss.sv.cmu.edu/ftsdata/> - I'm willing to invest more time to build on this. If there's interest, email me.

Questions

- How are those ReliefWeb FTS reports generated? Are they generated by machines or by humans?
 - I understand that they are generated by machines.
- If they're generated by machines, it may be easiest to tweak the the generation script. This would mean the response would be dynamically generated for any response, would be to amend the reports that are generated.
 - Agreed, but getting this 'tweak' done will probably not happen any time in the near future - I have asked the FTS developer about XML feeds and he said that they only had the one I listed below (that was relevant).....and, essentially, his time is already allotted so he will not be able to implement 'ad hoc' requests for things like this (i.e. it would have to be put into a project proposal, passed through the OCHA IT Governance board, etc, etc).
 - I will email the developer nonetheless again and see if he can help out (but I am not counting on it)
- If humans create them, are the reports always consistent?
 - Regarding the consistency: they are created out of a database - the variance between emergencies will be what clusters/sectors are included in the appeal. So, we cannot assume that the same number and name of clusters will be used.
- Finally, if it's possible to get a copyright waiver, it should be fairly easy for Sahana to consume these reports, extract the data & feed them out again as machine-readable documents. You'll be able to generate reports from our feeds.
 - I am really not sure on the copyright issues - and I could not find anything quickly on the site. Something to look into for me.
- I would recommend exporting Relief Web Excel sheets to CSV and pick require data from it to use in calculations, charts etc.
 - Yes - this was one question/suggestion that I received - make the data available in CSV. Unfortunately, data is only available in the formats that I listed. I will approach the FTS developer again, but I am not expecting any solution in the near future.
 - @Naveed - guys might be a stupid suggestion, but if we are going for a manual approach, then if we have the XLS file, I might be able to write a short program in C/Matlab that parses the file to come up with a CSV.
- By the way, I am planning to do something similar and am currently going through resource building phase (Kashif)
 - Will be interested to see what you come up with

FYI about past work on this topic from B. Tomaszewski:

In 2008, I worked as a consultant for OCHA/Relief web building a system to dynamically chart and map FTS data for display in Google Earth. This work could be leveraged for Pakistan and beyond.

The work was documented in the following paper I wrote with Lorant Czárán for the 2009

ISCRAM conference:

Tomaszewski, B. and Czárán, L. (2009) 'Geographically Visualizing Consolidated Appeal Process (CAP) Information', Proceedings of the 6th International ISCRAM Conference, pp. 1-6.

Paper Link:

http://www.iscram.org/ISCRAM2009/papers/Contributions/152_Geographically%20Visualizing%20Consolidated%20Appeal_Tomaszewski2009.pdf

This link is an example of KML produced by the system I developed that charts and maps FTS data:

http://www.ungiwg.org/RWVacancyGeoView/CAP_appeal_mapping_demo.kml

Note that this KML has 2 viewing levels – 1 level showing Global Appeal information and a second that shows individual CAP projects for the Sudan which was the test country we used at the time.

See the ISCRAM paper for further explanations about what you are seeing in the KML.

Some misc points about the system and past work:

- System uses Google Charts
- System will geocode CAP project titles to dynamically create a map of appeals
- Data was manually converted over from FTS spreadsheets into a RDMS so that data could then be processed to create the KML output. Engineering the data out of the FTS spreadsheets is of course the big challenge still. A completely automated system will be very difficult and would still require some human interaction
- The system I developed is written all in Java and could easily be re-purposed into a webservice for both input and output of FTS data or into whatever technology(s) is needed.

several prototypes using a mix of technologies (mostly Flex and Google) for ways to visualize FTS data with other variables:

<http://www.ungiwg.org/RWVacancyGeoView/CAPViz/bin/CAPViz.html>

http://www.ungiwg.org/RWVacancyGeoView/CAPViz/cap_motion_chart.html

http://www.ungiwg.org/RWVacancyGeoView/CAPViz/bin/sector_funding_requests.html

http://www.ungiwg.org/RWVacancyGeoView/CAPViz/cap_myanmar_timeline.html

Please add details here - so that people can see what is being done. I personally like to think of a Google Doc as a user friendly wiki :-)

Part 2 - World map of consolidate & flash appeal

UPDATE: Google and FTS have worked together in the past to work on something similar. I am not sure on the status of the work so I would not proceed on the below until clarity from FTS was given.

For this portion, I am much more open to creative solutions. Still, whatever it might be, it will need to be 'open', flexible, extremely easy to use - so easy that a 'non-techie' can make it work.

Request: generate a world map that shows points of all the appeals in a given year. Details of the given appeal would be displayed when the respective point is clicked on.

1. Show a combination of all the Consolidate Appeals, the Flash Appeals, and the Other Appeals (each type would have a different icon color)
2. [List of 2010 Appeals](#) (some of the 2010 summary tables are displayed at the top of the page)
3. When clicking on the given point, ideally one would see:
 1. Appeal Name / Date
 2. General Numbers: Total Amount of Appeal, Current Funding Level (and percentage), Amount still required (and percentage)
 3. Chart - pulled from Part 1
4. Map must be embeddable on other websites

Global Funding Tables

- If inclined, it would be very nice to show a similar chart/table as requested in Part 1 for the overall Global Funding.
- Global Funding tables: <http://fts.unocha.org/pageloader.aspx?page=emerg-globalOverview&year=2010>
- The chart would so the data corresponding the the year shown on the map.

What has happened so far:

- Uploaded and georeferenced: <http://finder.geocommons.com/overlays/61587>